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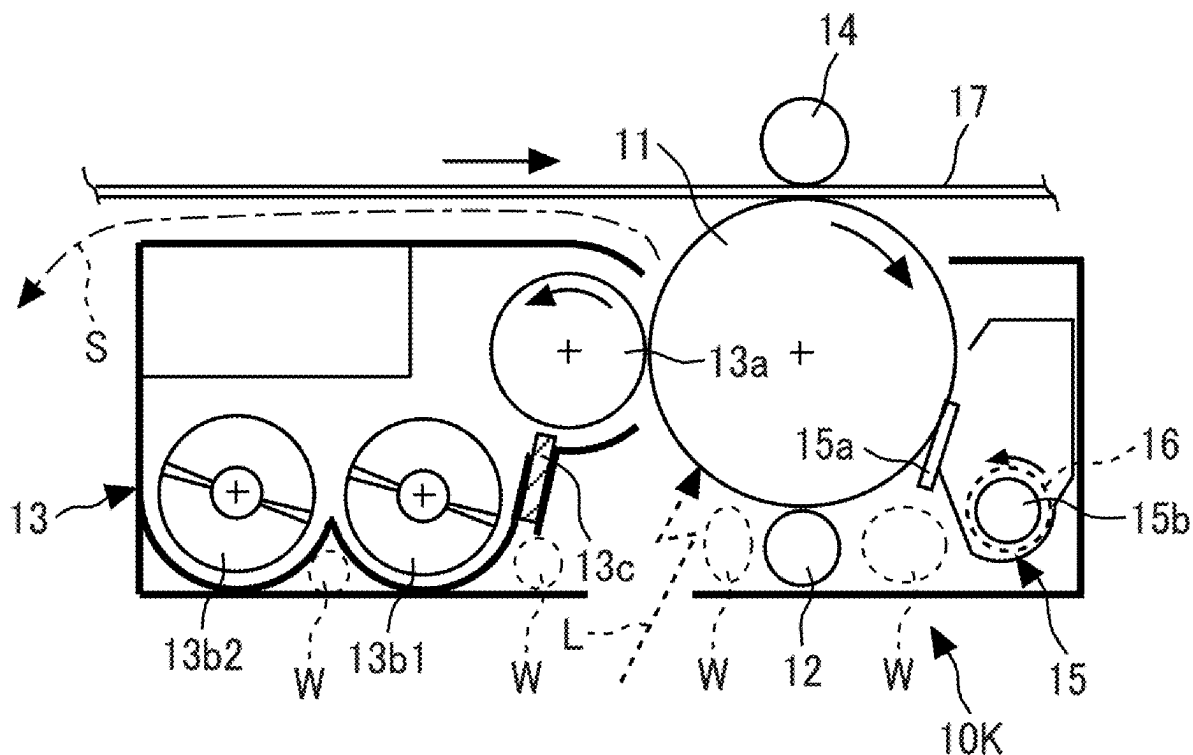


FIG. 1

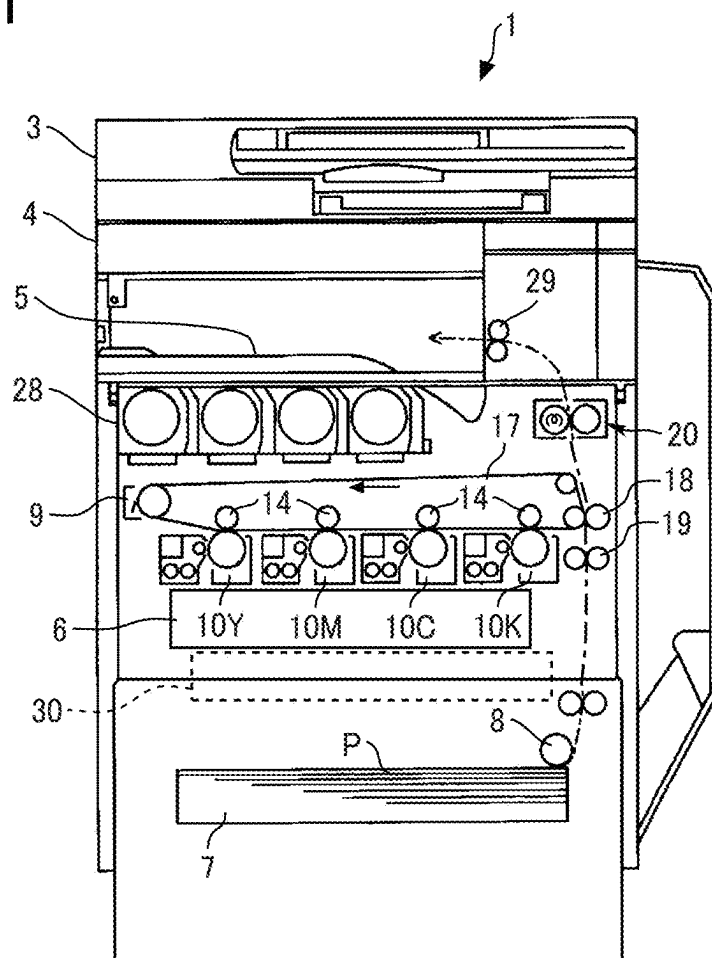


FIG. 2

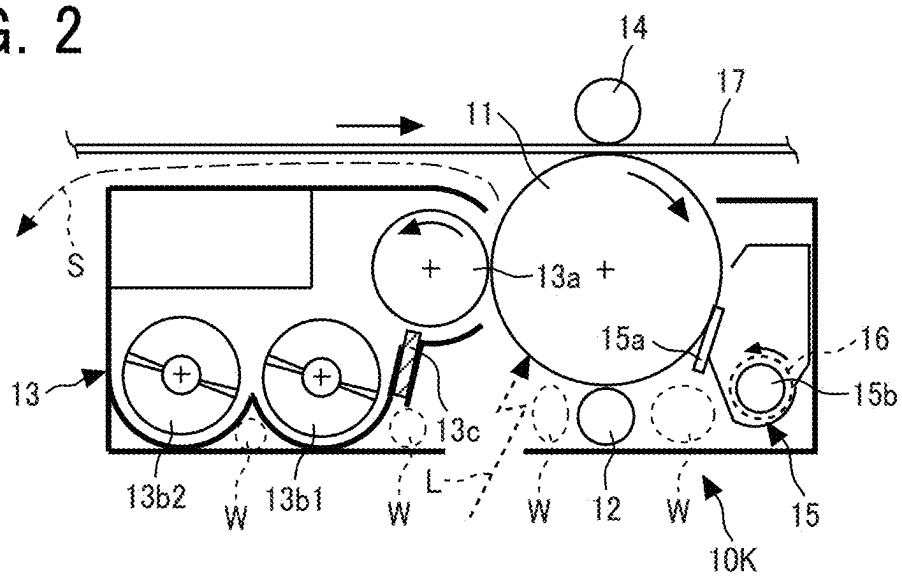


FIG. 3A

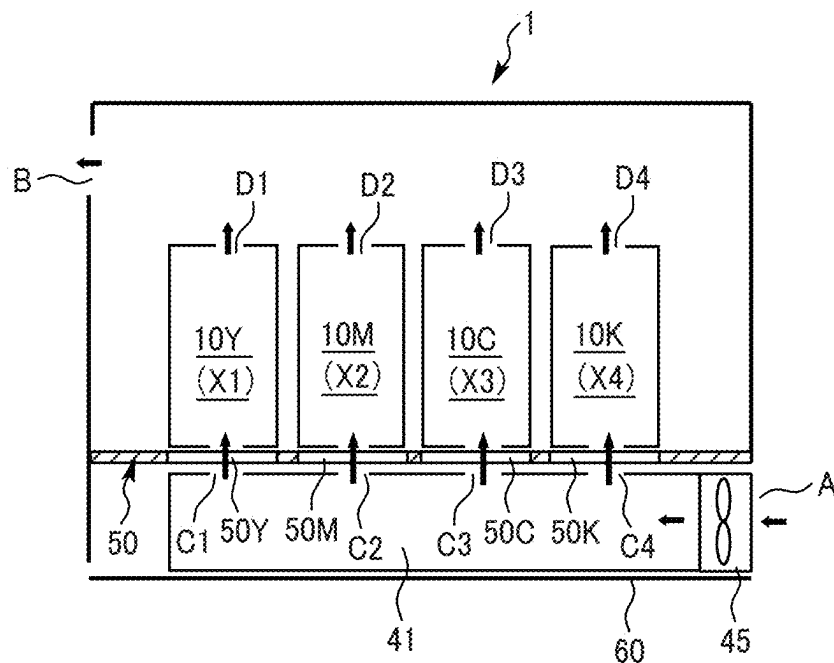


FIG. 3B

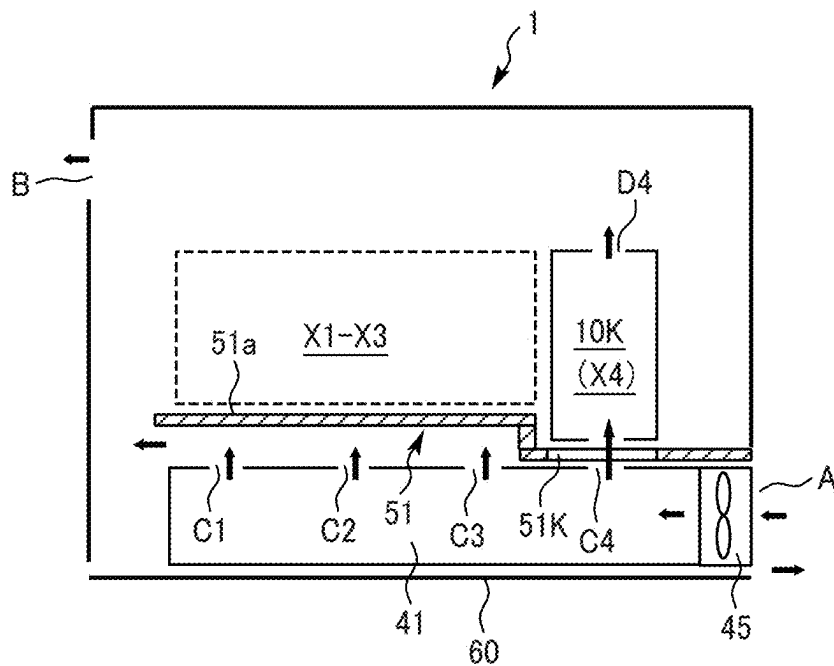


FIG. 4A

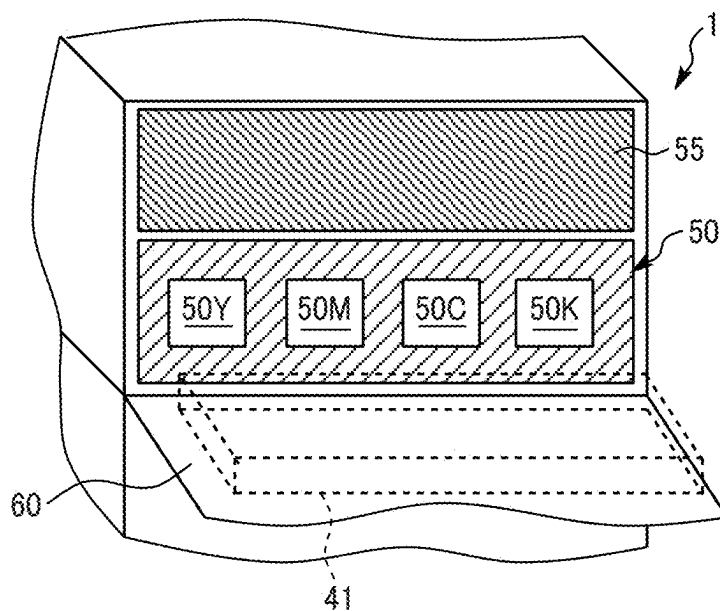


FIG. 4B

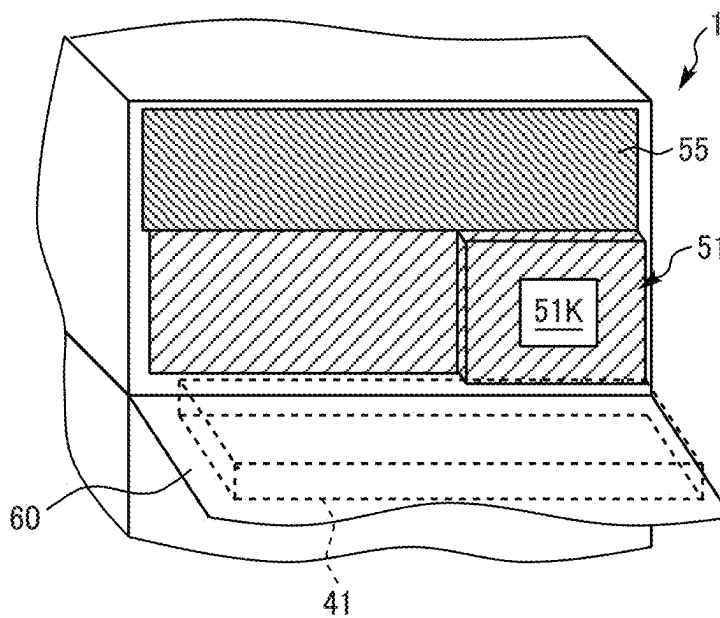


FIG. 5A

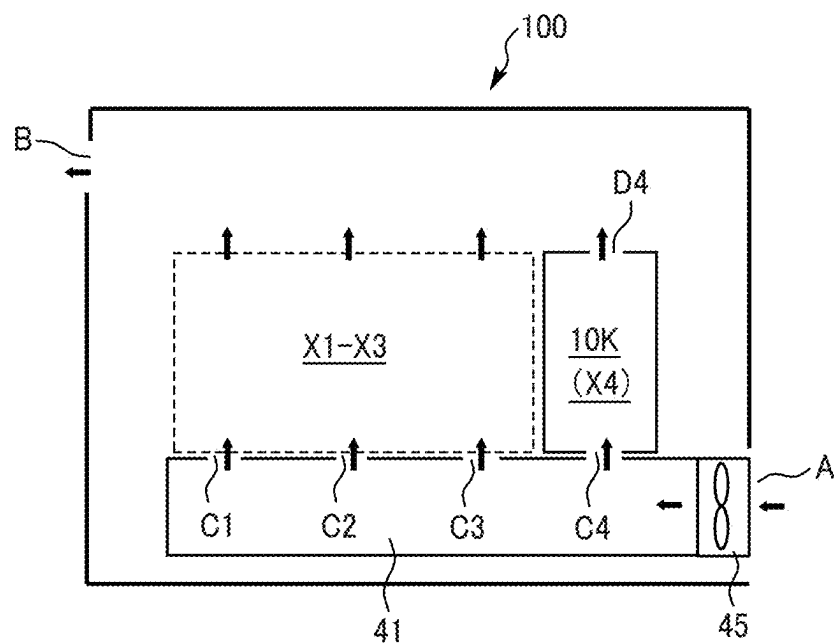


FIG. 5B

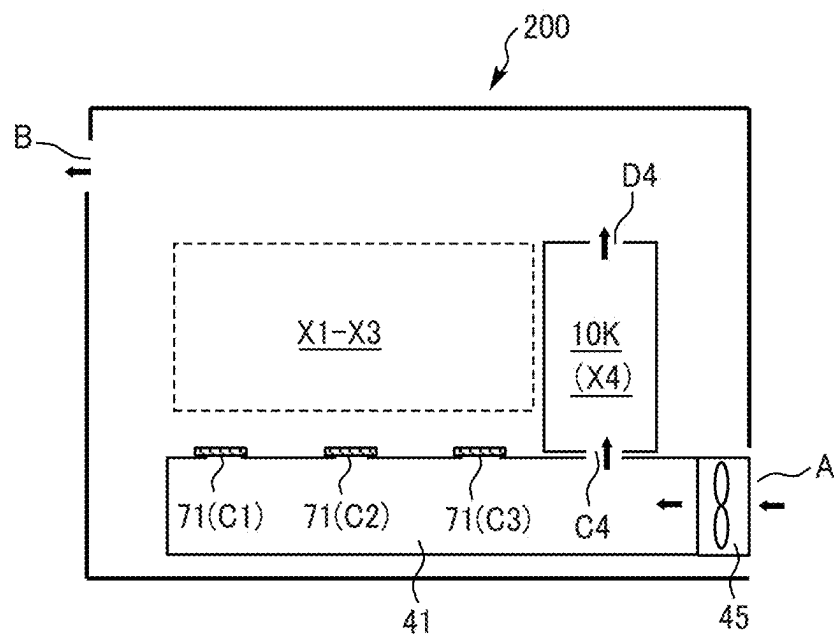


FIG. 6A

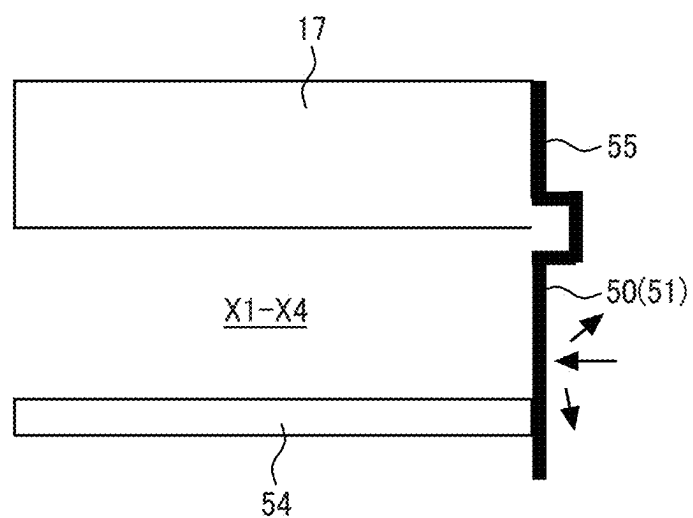


FIG. 6B

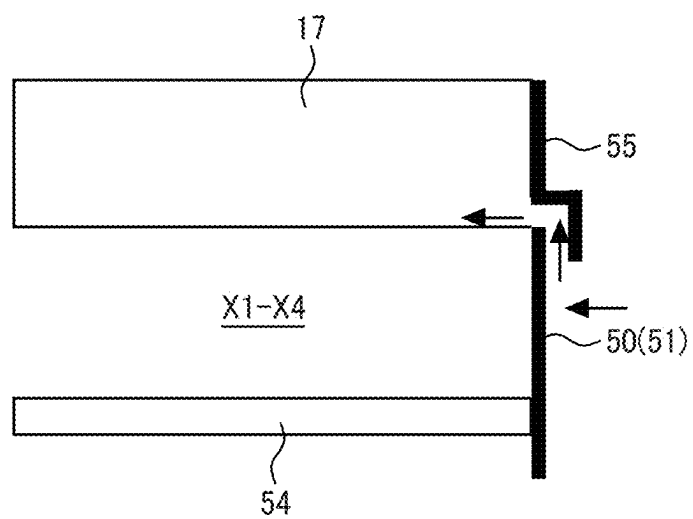


IMAGE FORMING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2024-031330, filed on Mar. 1, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] Embodiments of the present disclosure relate to an image forming apparatus such as a copying machine, a printer, a facsimile machine, or a multifunction peripheral thereof.

Related Art

[0003] In an image forming apparatus such as a copier or a printer, a technique is known that can perform image forming (printing) in a state where only one image forming unit is installed and the other image forming units are not installed in a space where a plurality of image forming units can be installed.

SUMMARY

[0004] In an embodiment of the present disclosure, an image forming apparatus includes a body, a plurality of image forming units, a plurality of stations, a plurality of air supply ports, an air supply duct, a full-specification inner cover, and a specified-specification inner cover. The plurality of image forming units are detachably attached in the plurality of stations. The plurality of air supply ports allow air supply to the plurality of stations. The air supply duct causes air taken in from outside of the body of the image forming apparatus to flow toward the plurality of air supply ports. The full-specification inner cover is installed in the body of the image forming apparatus when image formation is performed in a full specification state in which all of the plurality of image forming units are installed in the plurality of stations, allows air supply from all of the plurality of air supply ports to all of the plurality of stations, and allows attachment and detachment of the plurality of image forming units to and from the plurality of stations. The specified-specification inner cover is installed in the body of the image forming apparatus when image formation is performed in a specified specification state in which a station to be used, in which an image forming unit is installed, and an unused station, in which no image forming unit is installed, are present among the plurality of stations, allows air supply from an air supply port corresponding to the station to be used, among the plurality of air supply ports, to the station to be used, prevents air supply to the unused station via an air supply port other than the air supply port corresponding to the station to be used, and allows attachment and detachment of only the image forming unit to and from the station to be used.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages

and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0006] FIG. 1 is a diagram illustrating an overall configuration of an image forming apparatus according to an embodiment of the present disclosure;

[0007] FIG. 2 is a view of an image forming unit;

[0008] FIG. 3A is a top view of an image forming apparatus in a full specification state, illustrating flows of air therein;

[0009] FIG. 3B is a top view of the image forming apparatus of FIG. 3A in a specified specification state, illustrating flows of air therein;

[0010] FIG. 4A is a perspective view of an image forming apparatus in which an exterior cover is opened in a full specification state;

[0011] FIG. 4B is a perspective view of the image forming apparatus of FIG. 4A in which the exterior cover is opened in a specified specification state;

[0012] FIG. 5A is a top view of flows of air in an image forming apparatus in a specified specification state as a comparative example, illustrating flows of air therein;

[0013] FIG. 5B is a top view of flows of air in an image forming apparatus in a specified specification state as a comparative example, illustrating flows of air therein;

[0014] FIG. 6A is a side view of an image forming apparatus in a specified specification state, illustrating flows of air therein; and

[0015] FIG. 6B is a side view of an image forming apparatus in a specified specification state, illustrating flows of air therein.

[0016] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

[0017] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

[0018] Referring now to the drawings, embodiments of the present disclosure are described below. Like reference signs are assigned to like elements or components and descriptions of those elements or components may be simplified or omitted. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0019] A description is given below of an overall configuration and operation of an image forming apparatus 1 with reference to FIG. 1. In FIG. 1, the image forming apparatus 1, which is illustrated as a color copier in the present embodiment, includes a document conveying device 3, a scanner 4 (document reading device), and a writing device 6 (exposure device). The document conveying device 3 conveys documents to the scanner 4. The scanner 4 scans the documents to read image data. The writing device 6 emits a laser beam based on input image data. The image forming

apparatus 1 also includes a sheet feeder 7, image forming units 10Y, 10M, 10C, and 10K, an intermediate transfer belt 17, and a secondary transfer roller 18. The sheet feeder 7 stores sheets P such as sheets of paper. The image forming units 10Y, 10M, 10C, and 10K are image forming units to form toner images of yellow, magenta, cyan, and black, respectively. The toner images of multiple colors are transferred and superimposed one on another onto the intermediate transfer belt 17. The secondary transfer roller 18 transfers the toner images on the intermediate transfer belt 17 onto the sheet P. The image forming apparatus 1 further includes a fixing device 20, toner containers 28, and a waste-toner collection container 30. The fixing device 20 fixes unfixed toner images on the sheet P. The toner containers 28 contain toners of respective colors to be supplied to developing devices 13 of the corresponding image forming units 10Y, 10M, 10C, and 10K (process cartridges). Waste-toner is collected in the waste-toner collection container 30.

[0020] Each of the image forming units 10Y, 10M, 10C, and 10K (the process cartridges) includes a photoconductor drum 11 (serving as an image bearer), a charging device 12, the developing device 13, and a cleaning device 15, which are integrated as a single unit as illustrated in FIG. 2. Each of the image forming units 10Y, 10M, 10C, and 10K, which is expendable, is replaced with a new one when depleted. Yellow, magenta, cyan, and black toner images are formed on the respective photoconductor drums 11 (serving as image bearers) in the image forming units 10Y, 10M, 10C, and 10K.

[0021] A description is given below of operations of the image forming apparatus 1 to form a normal color toner image. Conveying rollers of the document conveying device 3 convey a document on a document table onto an exposure glass of the scanner 4. The scanner 4 optically scans the document on the exposure glass to read image data. The yellow, magenta, cyan, and black image data are transmitted to the writing device 6. The writing device 6 irradiates the photoconductor drums 11 of the corresponding image forming units 10Y, 10M, 10C, and 10K with laser beams L (exposure light) based on the yellow, magenta, cyan, and black image data, respectively.

[0022] Meanwhile, the four photoconductor drums 11 rotate clockwise as illustrated in FIGS. 1 and 2. With reference to FIG. 2, the charging device 12 (a charging roller) uniformly charges a surface of the photoconductor drum 11 at a position opposite the photoconductor drum 11 (a charging process). Thus, the surface of the photoconductor drum 11 is charged to a certain potential. Subsequently, the surface of the photoconductor drum 11 thus charged reaches a position where the surface of the photoconductor drum 11 is irradiated with the laser beam L. The writing device 6 emits, from a light source, the laser beams L for respective colors according to the image data of respective colors. The laser beams L are reflected by a polygon mirror and transmitted through multiple lenses. The laser beams L transmitted through the multiple lenses pass through different optical paths for the different color components of yellow, magenta, cyan, and black (an exposure process).

[0023] The laser beam L corresponding to the yellow image data is irradiated to the surface of the photoconductor drum 11 of the first image forming unit 10Y from the left in FIG. 1, to form an electrostatic latent image for yellow thereon. Thus, an electrostatic latent image corresponding to

the yellow image data is formed on the photoconductor drum 11 charged by the charging device 12. Similarly, the laser beam L corresponding to the cyan image data is irradiated to the surface of the photoconductor drum 11 of the second image forming unit 10C from the left in FIG. 1, to form an electrostatic latent image for cyan thereon. The laser beam L corresponding to the magenta image data is irradiated to the surface of the photoconductor drum 11 of the third image forming unit 10M from the left in FIG. 1, to form an electrostatic latent image for magenta thereon. The laser beam L corresponding to the black image data is irradiated to the surface of the photoconductor drum 11 of the fourth image forming unit 10K from the left in FIG. 1, to form an electrostatic latent image for black thereon.

[0024] Then, the surface of the photoconductor drum 11 bearing the electrostatic latent image for each color reaches the position opposite the developing device 13 (see FIG. 2). The developing device 13 supplies toner of each color onto the surface of the photoconductor drum 11 and develops the electrostatic latent image on the photoconductor drum 11 into a toner image (a development process). Subsequently, the surface of the photoconductor drum 11 after the development process reaches a position opposite the intermediate transfer belt 17 (intermediate transferor) as image bearer. Primary transfer rollers 14 are disposed at the positions where the surfaces of the photoconductor drums 11 face the intermediate transfer belt 17 such that the primary transfer rollers 14 contact an inner circumferential surface of the intermediate transfer belt 17. At the positions of the primary transfer rollers 14, the toner images on the photoconductor drums 11 are sequentially transferred and superimposed onto the intermediate transfer belt 17, forming a multicolor toner image thereon (a primary transfer process).

[0025] After the primary transfer process, the surface of the photoconductor drum 11 reaches the position opposite the cleaning device 15 (see FIG. 2). The cleaning device 15 collects untransferred toner remaining on the photoconductor drum 11 (a cleaning process). Then, the surface of the photoconductor drum 11 passes through a charge elimination device to complete a series of image forming processes performed on the photoconductor drum 11.

[0026] Meanwhile, the surface of the intermediate transfer belt 17, onto which the single-color toner images on the photoconductor drums 11 are transferred and superimposed, moves in a direction indicated by an arrow in FIG. 1 and reaches a position opposite a secondary transfer roller 18. The secondary transfer roller 18 secondarily transfers the multicolor toner image on the intermediate transfer belt 17 onto the sheet P (a secondary transfer process). After the secondary transfer process, the surface of the intermediate transfer belt 17 reaches a position opposite an intermediate transfer belt cleaner 9 (a cleaning device for the intermediate transfer belt 17). The intermediate transfer belt cleaner 9 collects the untransferred toner on the intermediate transfer belt 17 to complete a series of transfer processes on the intermediate transfer belt 17.

[0027] The sheet P is conveyed from the sheet feeder 7 to the position of the secondary transfer roller 18 via, for example, a sheet conveyance guide and a registration roller pair 19. More specifically, a feed roller 8 feeds the sheet P from the sheet feeder 7 that stores a stack of sheets P, and the sheet P is then guided by the sheet conveyance guide to the registration roller pair 19. The sheet P that has reached the registration roller pair 19 is conveyed toward the position of

the secondary transfer roller **18** so that the sheet P coincides with the arrival of the multicolor toner image on the intermediate transfer belt **17**.

[0028] Subsequently, the sheet P, onto which the multicolor toner image is transferred, is conveyed to a fixing device **20**. The fixing device **20** includes a fixing roller and a pressure roller pressing against each other. In a nip between the fixing roller and the pressure roller, the multicolor toner image is fixed on the sheet P. After the fixing process, an output roller pair **29** ejects the sheet P as an output image to the exterior of a body of the image forming apparatus **1**, and the ejected sheets P are stacked on an output tray **5** to complete a series of image forming processes.

[0029] With reference to FIG. 2, a description is given below of the image forming units **10Y**, **10M**, **10C**, and **10K** of the image forming apparatus **1** in detail. FIG. 2 is a diagram illustrating a configuration of the image forming unit **10K** for black. Each of the other three image forming units **10Y**, **10M**, and **10C** has substantially the same configuration as the image forming unit **10K** for black except for the color of toner used in the image forming process, and thus drawings and descriptions thereof are omitted to avoid redundancy.

[0030] As illustrated in FIG. 2, the image forming unit **10K** is a single unit that includes the photoconductor drum **11** as the image bearer, the charging device **12** to charge the photoconductor drum **11**, the developing device **13** to develop an electrostatic latent image on the photoconductor drum **11**, and the cleaning device **15** to remove untransferred toner from the photoconductor drum **11** in a casing of the image forming unit **10K**.

[0031] The photoconductor drum **11** is an organic photoconductor designed to be charged with a negative polarity and includes a photosensitive layer formed on a drum-shaped conductive support. The charging device **12** is a charging roller including a conductive core and an elastic layer of moderate resistivity overlaid on the conductive core. A power supply applies a specified voltage to the charging device **12** (charging roller). Thus, the charging device **12** uniformly charges the surface of the photoconductor drum **11** facing the charging device **12**.

[0032] The developing device **13** includes a developing roller **13a** disposed opposite the photoconductor drum **11**, a first conveying screw **13b1** disposed opposite the developing roller **13a**, a second conveying screw **13b2** disposed opposite the first conveying screw **13b1** via a partition, and a doctor blade **13c** disposed opposite the developing roller **13a**. The developing roller **13a** includes multiple magnets and a sleeve that rotates around the magnets. The magnets are stationary and generate magnetic poles around the circumferential surface of the developing roller **13a**. The magnets generate a plurality of magnetic poles on the developing roller **13a** (sleeve) to bear developer on the developing roller **13a**. The developing device **13** stores two-component developer including carrier and toner.

[0033] The cleaning device **15** includes a cleaning blade **15a** that contacts the photoconductor drum **11** and a conveying screw **15b** (a conveyance tube **16**) that conveys the untransferred toner collected in the cleaning device **15** toward a waste-toner conveying device as waste toner. For example, the cleaning blade **15a** is made of rubber, such as urethane rubber, and contacts the surface of the photoconductor drum **11** at a specified angle with a specified pressure. With such a configuration, substances such as the untrans-

ferred toner adhering to the photoconductor drum **11** are mechanically scraped off and collected in the cleaning device **15**. The untransferred toner collected in the cleaning device **15** is conveyed to the waste-toner conveying device via the conveyance tube **16** in which the conveying screw **15b** is disposed and conveyed to the waste-toner collection container **30** by the waste-toner conveying device. The conveyed untransferred toner is collected in the waste-toner collection container **30** as the waste toner. In addition to the untransferred toner, substances adhering to the photoconductor drum **11** or the intermediate transfer belt **17** include, for example, paper dust resulting from the sheet P, discharge products generated on the photoconductor drum **11** during discharge by the charging device **12**, and additives to the toner. In the present specification, such substances are collectively referred to as the “untransferred toner.”

[0034] The image forming processes, described above, are described in further detail below with reference to FIG. 2. The developing roller **13a** rotates in a direction (counterclockwise) indicated by an arrow in FIG. 2. In the developing device **13**, as the first conveying screw **13b1** and the second conveying screw **13b2** arranged via the partition rotate, the developer is circulated in the longitudinal direction of the developing device **13**, while being stirred and mixed with toner supplied from the toner container **28** by a toner supply device. The longitudinal direction of the developing device **13** is perpendicular to the plane on which FIG. 2 is illustrated.

[0035] Thus, the toner is triboelectrically charged and attracted to the carrier. The toner is borne on the developing roller **13a** together with the carrier. The developer borne on the developing roller **13a** reaches a position opposite the doctor blade **13c**. After having been adjusted to an appropriate amount at the position of the doctor blade **13c**, the developer on the developing roller **13a** then reaches an opposing position to the photoconductor drum **11** (i.e., a development area). In the development area, the toner in the developer adheres to the electrostatic latent image formed on the surface of the photoconductor drum **11**. The toner adheres to the electrostatic latent image (i.e., the toner image is formed) by a development electric field formed by a potential difference (i.e., a developing potential) between a latent image potential (i.e., an exposure potential) of an image area irradiated with the laser beam L and a development bias applied to the developing roller **13a**. Subsequently, most of the toner attached to the photoconductor drum **11** in the development process is transferred onto the intermediate transfer belt **17**. The untransferred toner remained on the surface of the photoconductor drum **11** is collected in the cleaning device **15** by the cleaning blade **15a**.

[0036] The configuration and operation of the image forming apparatus **1** according to the present embodiment are described in further detail below. As described above with reference to FIG. 1, the plurality of image forming units **10Y**, **10M**, **10C**, and **10K** are removably installed in the image forming apparatus **1**. Specifically, the plurality of image forming units are four image forming units, that is, one image forming unit **10K** for black and three image forming units **10Y**, **10M**, and **10C** for colors. With reference to FIGS. 3A and 4A, the image forming apparatus **1** includes a plurality of stations X1, X2, X3, and X4 (installation sections) to which the plurality of image forming units **10Y**, **10M**, **10C**, and **10K** are removably installed, respectively.

Specifically, the four image forming units **10Y**, **10M**, **10C**, and **10K** are attached to or detached from the four stations **X1**, **X2**, **X3**, and **X4**, respectively. When the image forming units **10Y**, **10M**, **10C**, and **10K** are not installed, the four stations **X1**, **X2**, **X3**, and **X4** are spaces that are entirely open.

[0037] With reference to FIG. 3A, the image forming apparatus **1** includes a flow passage through which air (outside air) taken in from the outside through an inlet port **A** is caused to flow into the body of the image forming apparatus **1** and is exhausted to the outside of the body of the image forming apparatus **1** through an outlet port **B**. In other words, flow passages of air (gas) as indicated by black arrows in FIG. 3A are formed in the image forming apparatus **1**. The flow passages are typically used to cool the image forming units **10Y**, **10M**, **10C**, and **10K**.

[0038] The image forming apparatus **1** (flow passage) is provided with, for example, a plurality of (four) air supply ports **C1**, **C2**, **C3**, and **C4**, an air supply duct **41**, and an air supply fan **45**. The plurality of air supply ports **C1**, **C2**, **C3**, and **C4** enable air supply to the plurality of stations **X1**, **X2**, **X3**, and **X4**, respectively. The air supply duct **41** is for allowing air taken in from the outside of the body of the image forming apparatus **1** via the inlet port **A** to flow toward the plurality of air supply ports **C1**, **C2**, **C3**, and **C4**. Specifically, in the present embodiment, the air supply duct **41** has the four air supply ports **C1**, **C2**, **C3**, and **C4** at positions facing the front sides (the lower sides in FIG. 3A and the front sides in the direction perpendicular to the plane on which FIG. 1 is illustrated) of the four image forming units **10Y**, **10M**, **10C**, and **10K**, respectively. The air supply duct **41** is provided with the air supply fan **45** upstream (on the right side in FIG. 3A) from the plurality of air supply ports **C1**, **C2**, **C3**, and **C4** in the air flow direction. When the air supply fan **45** is driven, the outside air is actively taken into the air supply duct **41** through the inlet port **A**.

[0039] With reference to FIG. 4A, in the present embodiment, an exterior cover **60** (an opening-and-closing cover) that can open the inside of the image forming apparatus **1** is disposed on the front side (the front side on which an operator such as a user performs typical operations) of the body of the image forming apparatus **1**. When the exterior cover **60** is opened (in the state of FIG. 4A), a full-specification inner cover **50** (inner cover) having insertion ports **50Y**, **50M**, **50C**, and **50K** for attachment to and detachment from the image forming units **10Y**, **10M**, **10C**, and **10K** is exposed. In the present embodiment, the air supply duct **41** (which is moved in accordance with an opening-and-closing operation of the exterior cover **60**) is disposed inside the exterior cover **60**. When the exterior cover **60** is closed, the plurality of air supply ports **C1**, **C2**, **C3**, and **C4** of the air supply duct **41** face openings of the image forming units **10Y**, **10M**, **10C**, and **10K** via the insertion ports **50Y**, **50M**, **50C**, and **50K** of the full-specification inner cover **50**, respectively. Note that the positional relation of the full-specification inner cover **50**, the air supply duct **41**, and the stations **X1**, **X2**, **X3**, and **X4** is not limited to the positional relation illustrated in FIG. 3A.

[0040] With reference to FIG. 3A, the plurality of exhaust ports **D1**, **D2**, **D3**, and **D4** enable exhaust from the image forming units **10Y**, **10M**, **10C**, and **10K** installed in the plurality of stations **X1**, **X2**, **X3**, and **X4**, respectively. The air exhausted from the plurality of exhaust ports **D1**, **D2**, **D3**, and **D4** is exhausted to the outside of the body of the image

forming apparatus **1** via the outlet port **B**. Specifically, in the present embodiment, the four exhaust ports **D1**, **D2**, **D3**, and **D4** are formed on the rear side (the upper side in FIG. 3A and the rear side in the direction perpendicular to the plane on which FIG. 1 is illustrated) of the four image forming units **10Y**, **10M**, **10C**, and **10K**, respectively.

[0041] In the present embodiment, the air that has flowed into the stations **X1**, **X2**, **X3**, and **X4** from the air supply ports **D1**, **D2**, **D3**, and **D4** passes through the inside of the image forming units **10Y**, **10M**, **10C**, and **10K** (typically, spaces **W** surrounded by the dashed lines in FIG. 2) installed in the stations **X1**, **X2**, **X3**, and **X4**, and is exhausted from the exhaust ports **D1**, **D2**, **D3**, and **D4**, respectively. In other words, the air is released from the front side to the rear side in the image forming units **10Y**, **10M**, **10C**, and **10K**. Such a configuration can enhance the cooling performance of the image forming units **10Y**, **10M**, **10C**, and **10K**.

[0042] The image forming apparatus **1** according to the present embodiment can perform image formation (printing) in a “full specification state” in which all of the plurality of (four) image forming units **10Y**, **10M**, **10C**, and **10K** are installed in the plurality of (four) stations **X1**, **X2**, **X3**, and **X4** as illustrated in FIGS. 1, 3A, and 4A, or in a “specified specification state” in which the station **X4** to be used, in which the image forming unit **10K** is installed, and the unused stations **X1**, **X2**, and **X3**, in which the image forming units **10Y**, **10M**, and **10C** are not installed, are present among the plurality of (four) stations **X1**, **X2**, **X3**, and **X4** as illustrated in FIGS. 3B and 4B. The image forming apparatus **1** can switch the two states described above. In the present embodiment, the “specified specification state” is a state in which monochrome image formation (monochrome printing) can be performed using only the image forming unit **10K** for black. Accordingly, the image forming apparatus **1** according to the present embodiment can be used by a user as a full-color image forming apparatus using four colors (**Y**, **M**, **C**, and **K**) or as a monochrome image forming apparatus using only black. Such a configuration can reduce the cost required for design and manufacturing compared to a case where a full-color image forming apparatus and a monochrome image forming apparatus are separately manufactured. Even when a user desires to change the image forming apparatus from a full-color image forming apparatus to a monochrome image forming apparatus or from a monochrome image forming apparatus to a full-color image forming apparatus, such a configuration can meet the desire of the user without much cost.

[0043] In the present embodiment, as illustrated in FIG. 3A, when image formation is performed in the “full specification state” (when the image forming apparatus **1** is used as the full-color image forming apparatus), air is supplied from all of the plurality of air supply ports **C1**, **C2**, **C3**, and **C4** to all of the plurality of stations **X1**, **X2**, **X3**, and **X4**. When image formation is performed in the “full specification state” (when the image forming apparatus **1** is used as the full-color image forming apparatus), the exhaust is performed from all of the plurality of stations **X1**, **X2**, **X3**, and **X4** to an exhaust duct **42** via the plurality of exhaust ports **C1**, **C2**, **C3**, and **C4**. In other words, in the “full specification state,” air flows are formed as indicated by the black arrows in FIG. 3A. As a result, as described above, such a configuration can enhance the cooling performance (the performance of restricting the temperature rise) of the image forming units **10Y**, **10M**, **10C**, and **10K**.

[0044] With reference to FIGS. 3A and 4A, when image formation is performed in the “full specification state” (when the full-color image forming apparatus 1 is used), the full-specification inner cover 50 is installed inside the exterior cover 60. The full-specification inner cover 50 allows air supply from all of the plurality of air supply ports C1, C2, C3, and C4 to all of the plurality of stations X1, X2, X3, and X4 and allows attachment and detachment of the plurality of image forming units 10Y, 10M, 10C, and 10K to and from the plurality of stations X1, X2, X3, and X4. In the full-specification inner cover 50, as described above, the plurality of insertion ports 50Y, 50M, 50C, and 50K for attachment and detachment of the plurality of image forming units 10Y, 10M, 10C, and 10K are formed so as not to hinder the air supply from the plurality of air supply ports C1, C2, C3, and C4 of the air supply duct 41 to the image forming units 10Y, 10M, 10C, and 10K. The full-specification inner cover 50 is removably installed to a housing of the body of the image forming apparatus 1 by a relatively simple fixing way such as screwing and can be attached to and detached from the housing of the body of the image forming apparatus 1 in a state where the exterior cover 60 is open.

[0045] On the other hand, as illustrated in FIG. 3B, when image formation is performed in the “specified specification state” (when the image forming apparatus 1 is used as the monochrome image forming apparatus), air is supplied to the station X4 to be used (the image forming unit 10K for black) from the air supply port C4 corresponding to the station X4 to be used among the plurality of (four) air supply ports C1, C2, C3, and C4, and air is not supplied to the unused stations X1, X2, and X3 (the image forming units 10Y, 10M, and 10C for yellow, magenta, and cyan) via the other air supply ports C1, C2, and C3.

[0046] Specifically, in the “specified specification state” (when the image forming apparatus 1 is used as a monochrome image forming apparatus), a specified-specification inner cover 51 is installed inside the exterior cover 60. In other words, when the image forming apparatus 1 is changed from the full specification state to the specified specification state, the full-specification inner cover 50 is removed and the specified-specification inner cover 51 is installed. On the other hand, when the image forming apparatus 1 is changed from the specified specification state to the full specification state, the specified-specification inner cover 51 is removed and the full-specification inner cover 50 is installed.

[0047] The specified-specification inner cover 51 enables air supply from the air supply port C4 corresponding to the station X4 to be used (image forming unit 10K for black) among the plurality of air supply ports C1, C2, C3, and C4. The specified-specification inner cover 51 prevents air supply to the unused stations X1, X2, and X3 through the other air supply ports C1, C2, and C3, and enables only the image forming units 10Y, 10M, and 10C for yellow, magenta, and cyan to be attachable and detachable to and from the unused stations X1, X2, and X3. Specifically, in the specified specification state, no insertion ports for attachment and detachment of the image forming units 10Y, 10M, and 10C for yellow, magenta, and cyan are necessary. For this reason, only an insertion port 51K for attachment and detachment of the image forming unit 10K for black is formed in the specified-specification inner cover 51. As described above, the air supply duct 41 is disposed inside the exterior cover 60. When the exterior cover 60 is closed, one air supply port C4 (an air supply port corresponding to the station X4 to be

used) in the air supply duct 41 faces the opening of the image forming unit 10K for black via the insertion port 51K of the specified-specification inner cover 51. The air supply duct 41 is used in common in both the full specification state and the specified specification state. The specified-specification inner cover 51 as described above is installed, so that an inconvenience that the image forming units 10Y, 10M, and 10C are erroneously set in the unused stations X1, X2, and X3 in the specified specification state can be prevented. The specified-specification inner cover 51 is removably installed in the housing of the body of the image forming apparatus 1 by a relatively simple fixing way such as screwing, and can perform attachment and detachment in a state where the exterior cover 60 is open.

[0048] As illustrated in FIG. 3B, the specified-specification inner cover 51 is provided with a partition 51a that functions as a shielding member that separates the unused stations X1, X2, and X3 and the three air supply ports C1, C2, and C3 of the air supply duct 41 with a gap therebetween. The partition 51a (shielding member) is formed such that one end (on the right side in FIG. 3B) contacts the image forming unit 10K with substantially no clearance and an upper end contacts an intermediate-transfer inner cover 55 (see FIGS. 4A, 4B, and 6A) with substantially no clearance, thereby separating a part or all of the spaces X1, X2, and X3 in which the three image forming units 10Y, 10M, and 10C for yellow, magenta, and cyan can be installed, when viewed from the front side (the side on which the air supply duct 41 is disposed). With such a configuration, air is supplied from the air supply duct 41 only to the station X4 (the station to be used) corresponding to the image forming unit 10K for black among the four stations X1, X2, X3, and X4, and air is not supplied from the air supply duct 41 to the other stations X1, X2, and X3 (unused stations).

[0049] In the present embodiment, the partition 51a (shielding member) is formed such that air that has not been supplied to the station X4 to be used (image forming unit 10K for black) from the air supply port C4 corresponding to the station X4 to be used (image forming unit 10K for black), among the air that has flowed into the air supply duct 41 from the outside of the body of the image forming apparatus 1, can be exhausted to a side away from the plurality of stations X1, X2, X3, and X4. The air that has not been supplied to the station X4 to be used may be exhausted to the outside of the body of the image forming apparatus 1. Accordingly, in the specified specification state, the air flows as indicated by the black arrows in FIG. 3B are formed as a whole. As a result, as described above, the cooling performance (the performance of restricting the temperature rise) of the image forming unit 10K for black is enhanced.

[0050] As described above, in the present embodiment, the change from the full specification state to the specified specification state and the change from the specified specification state to the full specification state can be achieved by a simple operation in which replacement of the full-specification inner cover 50 and the specified-specification inner cover 51 is performed from the front side (the lower side in FIGS. 3A and 3B and the side on which the exterior cover 60 is installed) where an operator operates.

[0051] As illustrated in FIG. 3A, in the full specification state, a flow passage through which air passes from the front side to the rear side is formed in each of the four image forming units 10Y, 10M, 10C, and 10K, and thus the cooling

performance for all the image forming units **10Y**, **10M**, **10C**, and **10K** can be substantially uniformly enhanced.

[0052] As illustrated in FIG. 3B, in the specified specification state, a flow passage through which air passes from the front side to the rear side is formed only in the image forming unit **10K** for black (the station **X4** to be used), and such flow passages are not formed in the other unused stations **X1**, **X2**, and **X3**. Thus, the cooling performance for the image forming unit **X1** for black (the station **X4** to be used) can be enhanced compared to a case where such flow passages are formed in the unused stations **X1**, **X2**, and **X3** as in an image forming apparatus **100** illustrated in FIG. 5A as a comparative example.

[0053] As illustrated in FIG. 3B, in the specified specification state, the partition **51a** is disposed as described above, so that a part of the outside air flowing into the air supply duct **41** is ejected to the outside of the air supply duct **41** without flowing into the three air supply ports **C1**, **C2**, and **C3**. Accordingly, as compared with a case where only the air supply port **C4** corresponding to the image forming unit **10K** for black (the station **X4** to be used) is opened and the air supply ports **C1**, **C2**, and **C3** of the unused stations **X1**, **X2**, and **X3** are sealed by sealing members **71** as in an image forming apparatus **200** illustrated in FIG. 5B as a comparative example, the amount of air supplied to the image forming unit **10K** for black (station **X4** to be used) is not excessively increased and is appropriately adjusted. As a result, an inconvenience that toner scattering generated in the developing device **13** is diffused due to the excessive amount of air is less likely to occur.

[0054] With reference to FIGS. 3A and 3B, any one of the full-specification inner cover **50** and the specified-specification inner cover **51** is installed with a clearance on the inner side (the inner side and the upper side in FIGS. 3A and 3B) with respect to the exterior cover **60** in the body of the image forming apparatus **1**. Specifically, in the image forming apparatus **1**, the air supply duct **41** and the full-specification inner cover **50** or the air supply duct **41** and the specified-specification inner cover **51** do not completely contact with each other, but a slight clearance is provided therebetween. Thus, even if air unexpectedly leaks from the air supply duct **41**, the air can be ejected to the outside of the image forming apparatus **1** through the clearance. As a result, the amount of air to be supplied to the image forming unit can be optimized without control adjustment or replacement of the air supply fan **45**.

[0055] As illustrated in FIGS. 6A and 6B, an intermediate-transfer-belt device including the intermediate transfer belt **17** is installed above the plurality of stations **X1**, **X2**, **X3**, and **X4**. As illustrated in FIGS. 4A, 4B, 6A, and 6B, the intermediate-transfer inner cover **55** is installed to be interposed between the intermediate-transfer-belt device (intermediate transfer belt **17**) and the exterior cover **60** (see FIGS. 4A and 4B) on the front side (the right side in FIGS. 6A and 6B) of the body of the image forming apparatus **1**. As illustrated in FIG. 6A, in the present embodiment, any one of the full-specification inner cover **50** and the specified-specification inner cover **51** is in contact with the intermediate-transfer inner cover **55** so that no clearance from the intermediate-transfer inner cover **55** is formed in the vertical direction. With this configuration, as indicated by arrows in FIG. 6A, foreign matter is less likely to enter the intermediate-transfer-belt device (intermediate transfer belt **17**) together with the outside air. In contrast, as illustrated in

FIG. 6B, any one of the full-specification inner cover **50** and the specified-specification inner cover **51** may be installed so that a clearance from the intermediate-transfer inner cover **55** is formed in the vertical direction. In such a case, as indicated by arrows in FIG. 6B, outside air can be actively taken into the intermediate-transfer-belt device (intermediate transfer belt **17**), and thus the temperature rise of the intermediate-transfer-belt device (intermediate transfer belt **17**) can be reduced. In the present embodiment, as illustrated in FIGS. 6A and 6B, a rail **54** that guides the attachment and detachment of each of the plurality of image forming units **10Y**, **10M**, **10C**, and **10K** is disposed below corresponding one of the plurality of stations **X1**, **X2**, **X3**, and **X4**.

[0056] In the present embodiment, any one of the full-specification inner cover **50** and the specified-specification inner cover **51** is installed as one component (inner cover) to face the plurality of stations **X1**, **X2**, **X3**, and **X4**. In other words, each of the full-specification inner cover **50** and the specified-specification inner cover **51** is formed as one component. In contrast, at least one of the full-specification inner cover **50** and the specified-specification inner cover **51** may be installed to face the plurality of stations **X1**, **X2**, **X3**, and **X4** as one component (inner cover) in which a plurality of components (inner covers) are united. In other words, at least one of the full-specification inner cover **50** and the specified-specification inner cover **51** may be formed as one component (inner cover) in which a plurality of components (inner covers) are joined and united by screwing or snap-on clipping. Even in such a case, the functions of the full-specification inner cover **50** and the specified-specification inner cover **51** as described above are exhibited.

[0057] As described above, the image forming apparatus **1** according to the present embodiment includes the plurality of stations **X1**, **X2**, **X3**, and **X4** to and from which the plurality of image forming units **10Y**, **10M**, **10C**, and **10K** are attachable and detachable. The image forming apparatus **1** is provided with the plurality of air supply ports **C1**, **C2**, **C3**, and **C4** that allow air to be supplied to the plurality of stations **X1**, **X2**, **X3**, and **X4**, respectively, and the air supply duct **41** that allows air taken in from the outside of the image forming apparatus body **1** to flow toward the plurality of air supply ports **C1**, **C2**, **C3**, and **C4**. The image forming apparatus **1** includes the full-specification inner cover **50** that is disposed in the body of the image forming apparatus **1** when image formation is performed in the full specification state in which all of the plurality of image forming units **10Y**, **10M**, **10C**, and **10K** are installed in the plurality of stations **X1**, **X2**, **X3**, and **X4**. The full-specification inner cover **50** enables air supply from all of the plurality of air supply ports **C1**, **C2**, **C3**, and **C4** to all of the plurality of stations **X1**, **X2**, **X3**, and **X4** and enables attachment and detachment of the plurality of image forming units **10Y**, **10M**, **10C**, and **10K** to the plurality of stations **X1**, **X2**, **X3**, and **X4**, respectively. The image forming apparatus **1** further includes the specified-specification inner cover **51**. When image formation is performed in the specified specification state in which the station **X4** to be used in which the image forming unit **10K** is installed and the unused stations **X1**, **X2**, and **X3** in which the image forming units **10Y**, **10M**, and **10C** are not installed are present among the plurality of stations **X1**, **X2**, **X3**, and **X4**. The specified-specification inner cover **51** enables air supply from the air supply port **C4** corresponding to the station **X4** to be used among the plurality of air supply ports **C1**, **C2**, **C3**, and **C4**, prevents air

from being supplied to the unused stations X1, X2, and X3 via the other air supply ports C1, C2, and C3, and enables attachment and detachment of only the image forming unit 10K to and from the station X4 to be used. Such a configuration can easily change from the full specification state to the specified specification state and can easily change from the specified specification state to the full specification state. [0058] In the image forming apparatus 1 according to the present embodiment, the image forming units 10Y, 10M, 10C, and 10K of four colors (Y, M, C, and K) are installed in the full specification state and one image forming unit 10K is installed in the specified specification state. However, the number of image forming units installed in the full specification state and the number of image forming units installed in the specified specification state are not limited to those in the present embodiment. For example, the number of image forming units installed in the full specification state may be five, which includes four for full color (Y, M, C, and K) and additional one for clear color or infrared. Further, the number of image forming units installed in the specified specification state may be three for color (Y, M, and C). In the present disclosure, the shape of the air supply duct 41, the positions of the air supply ports C1, C2, C3, and C4, and the positions of the exhaust ports D1, D2, D3, and D4 are not limited to those in the present embodiment. In the present embodiment, as illustrated in FIGS. 4A and 4B, when the specification of the image forming apparatus 1 is changed, the full-specification inner cover 50 is replaced with the specified-specification inner cover 51. In contrast, the inner cover facing the image forming unit 10K for black (station X4) used in any specification may be used in common and fixed at all times, and the inner covers facing the image forming units 10Y, 10M, and 10C for yellow, magenta, and cyan (stations X1, X2, and X3) that are used or not used depending on the specification may be replaced with those corresponding to the respective specifications. Such cases can also provide substantially the same or similar effects as the effects described above.

[0059] Note that embodiments of the present disclosure are not limited to the above-described embodiments and it is apparent that the above-described embodiments can be appropriately modified within the scope of the technical idea of the present embodiment in addition to what is suggested in the above-described embodiments. Further, features of components of the embodiments, such as the number, the position, and the shape are not limited the embodiments and thus may be preferably set.

[0060] Aspects of the present disclosure may be, for example, combinations of first to ninth aspects as follows.

FIRST ASPECT

[0061] An image forming apparatus (e.g., the image forming apparatus 1) includes a plurality of stations (e.g., the stations X1, X2, X3, and X4), a plurality of air supply ports (e.g., the air supply ports C1, C2, C3, and C4), an air supply duct (e.g., the air supply duct 41), a full-specification inner cover (e.g., the full-specification inner cover 50), and a specified-specification inner cover (e.g., the specified-specification inner cover 51). A plurality of image forming units (e.g., the image forming units 10Y, 10M, 10C, and 10K) are attachable to and detachable from the plurality of stations. The plurality of air supply ports allow air supply to the plurality of stations. The air supply duct causes air taken in from outside of a body of the image forming apparatus to

flow toward the plurality of air supply ports. When image formation is performed in a full specification state in which all of the plurality of image forming units are installed in the plurality of stations, the full-specification inner cover is installed in the body of the image forming apparatus to allow air supply from all of the plurality of air supply ports to all of the plurality of stations and allow attachment and detachment of the plurality of image forming units to and from the plurality of stations. When image formation is performed in a specified specification state in which a station to be used, in which an image forming unit is installed, and an unused station, in which no image forming unit is installed, are present among the plurality of stations, the specified-specification inner cover is installed in the body of the image forming apparatus to allow air supply from an air supply port corresponding to the station to be used, among the plurality of air supply ports, to the station to be used, prevent air supply to the unused station via an air supply port other than the air supply port corresponding to the station to be used, and allow attachment and detachment of only the image forming unit to and from the station to be used.

SECOND ASPECT

[0062] The image forming apparatus (e.g., the image forming apparatus 1) according to the first aspect further includes an exterior cover (e.g., the exterior cover 60) that can open an inside of the body of the image forming apparatus and is installed on a front side where an operator operates. Any one of the full-specification inner cover (e.g., the full-specification inner cover 50) and the specified-specification inner cover (e.g., the specified-specification inner cover 51) is installed with a clearance on an inner side with respect to the exterior cover in the body of the image forming apparatus.

THIRD ASPECT

[0063] In the image forming apparatus (e.g., the image forming apparatus 1) according to the first or second aspect, any one of the full-specification inner cover (e.g., the full-specification inner cover 50) and the specified-specification inner cover (e.g., the specified-specification inner cover 51) is installed as one component or one component in which a plurality of components are united to face the plurality of stations (e.g., the stations X1, X2, X3, and X4).

FOURTH ASPECT

[0064] The image forming apparatus (e.g., the image forming apparatus 1) according to any one of the first to third aspects further includes an intermediate-transfer-belt device and an intermediate-transfer inner cover (e.g., the intermediate-transfer inner cover 55). The intermediate-transfer-belt device is disposed above the plurality of stations (e.g., the stations X1, X2, X3, and X4). The intermediate-transfer inner cover is disposed to be interposed between the intermediate-transfer-belt device and the intermediate-transfer inner cover on a front side of the body of the image forming apparatus. Any one of the full-specification inner cover (e.g., the full-specification inner cover 50) and the specified-specification inner cover (e.g., the specified-specification inner cover 51) contacts so that no clearance from the intermediate-transfer inner cover is formed in a vertical direction.

FIFTH ASPECT

[0065] The image forming apparatus (e.g., the image forming apparatus 1) according to any one of the first to third aspects further includes an intermediate-transfer-belt device and an intermediate-transfer inner cover (e.g., the intermediate-transfer inner cover 55). The intermediate-transfer-belt device is disposed above the plurality of stations (e.g., the stations X1, X2, X3, and X4). The intermediate-transfer inner cover is disposed to be interposed between the intermediate-transfer-belt device and the intermediate-transfer inner cover on a front side of the body of the image forming apparatus. Any one of the full-specification inner cover (e.g., the full-specification inner cover 50) and the specified-specification inner cover (e.g., the specified-specification inner cover 51) contacts so that a clearance from the intermediate-transfer inner cover is formed in a vertical direction.

SIXTH ASPECT

[0066] In the image forming apparatus (e.g., the image forming apparatus 1) according to any one of the first to fifth aspects, the specified-specification inner cover (e.g., the specified-specification inner cover 51) is formed so that air that has not been supplied to the station to be used from the air supply port corresponding to the station to be used, among the air that has been taken in the air supply duct (e.g., the air supply duct 41) from the outside of the body of the image forming apparatus, is exhausted to a side away from the plurality of stations (e.g., the stations X1, X2, X3, and X4) as it is.

SEVENTH ASPECT

[0067] In the image forming apparatus (e.g., the image forming apparatus 1) according to any one of the first to sixth aspects, air that has flowed into the station to be used from the air supply port corresponding to the station to be used passes through an inside of the image forming unit installed in the station to be used.

EIGHTH ASPECT

[0068] In the image forming apparatus (e.g., the image forming apparatus 1) according to any one of the first to seventh aspects, the air supply duct (e.g., the air supply duct 41) is provided with an air supply fan (e.g., the air supply fan 45) upstream from the plurality of air supply ports (e.g., the air supply ports C1, C2, C3, and C4) in an air flow direction.

NINTH ASPECT

[0069] In the image forming apparatus (e.g., the image forming apparatus 1) according to any one of the first to eighth aspects, the plurality of image forming units (e.g., the image forming units 10Y, 10M, 10C, and 10K) include an image forming unit for black and three image forming units for colors other than black. The specified specification state is a state in which monochrome image formation can be performed using only the image forming unit for black.

[0070] The above-described embodiments are illustrative and do not limit the present disclosure. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined

with each other and/or substituted for each other within the scope of the present disclosure.

1. An image forming apparatus comprising:

- a body;
- a plurality of image forming units;
- a plurality of stations in which the plurality of image forming units are detachably attached;
- a plurality of air supply ports to allow air supply to the plurality of stations;
- an air supply duct to cause air taken in from outside of the body of the image forming apparatus to flow toward the plurality of air supply ports;
- a full-specification inner cover installed in the body of the image forming apparatus when image formation is performed in a full specification state in which all of the plurality of image forming units are installed in the plurality of stations, the full-specification inner cover allowing air supply from all of the plurality of air supply ports to all of the plurality of stations and allowing attachment and detachment of the plurality of image forming units to and from the plurality of stations, and
- a specified-specification inner cover installed in the body of the image forming apparatus when image formation is performed in a specified specification state in which a station to be used, in which an image forming unit is installed, and an unused station, in which no image forming unit is installed, are present among the plurality of stations, the specified-specification inner cover allowing air supply from an air supply port corresponding to the station to be used, among the plurality of air supply ports, to the station to be used, preventing air supply to the unused station via an air supply port other than the air supply port corresponding to the station to be used, and allowing attachment and detachment of only the image forming unit to and from the station to be used.

2. The image forming apparatus according to claim 1, further comprising an exterior cover to open an inside of the body of the image forming apparatus and installed on a front side where an operator operates,

wherein any one of the full-specification inner cover and the specified-specification inner cover is installed with a clearance on an inner side with respect to the exterior cover in the body of the image forming apparatus.

3. The image forming apparatus according to claim 1, wherein any one of the full-specification inner cover and the specified-specification inner cover is installed, as one component or one united component in which a plurality of components are united, to face the plurality of stations.

4. The image forming apparatus according to claim 3, further comprising:

- an intermediate-transfer-belt device disposed above the plurality of stations; and
- an intermediate-transfer inner cover interposed between the intermediate-transfer-belt device and the intermediate-transfer inner cover on a front side of the body of the image forming apparatus,

wherein any one of the full-specification inner cover and the specified-specification inner cover contacts without any clearance from the intermediate-transfer inner cover in a vertical direction.

5. The image forming apparatus according to claim 3, further comprising:

an intermediate-transfer-belt device disposed above the plurality of stations; and

an intermediate-transfer inner cover interposed between the intermediate-transfer-belt device and the intermediate-transfer inner cover on a front side of the body of the image forming apparatus,

wherein any one of the full-specification inner cover and the specified-specification inner cover contacts with a clearance from the intermediate-transfer inner cover in a vertical direction.

6. The image forming apparatus according to claim 1,

wherein the specified-specification inner cover is formed so that air that has not been supplied to the station to be used from the air supply port corresponding to the station to be used, among the air that has been taken in the air supply duct from the outside of the body of the

image forming apparatus, is exhausted to a side away from the plurality of stations.

7. The image forming apparatus according to claim 1, wherein air that has flowed into the station to be used from the air supply port corresponding to the station to be used passes through an inside of the image forming unit installed in the station to be used.

8. The image forming apparatus according to claim 1, wherein the air supply duct is provided with an air supply fan upstream from the plurality of air supply ports in an air flow direction.

9. The image forming apparatus according to claim 1, wherein the plurality of image forming units include an image forming unit for black and three image forming units for colors other than black, and wherein the specified specification state is a state in which monochrome image formation is performed using only the image forming unit for black.

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